MBU DIGITAL LIBRARY · PRACTICE LEARNING RESOURCE

Artificial Intelligence

AI101 | Suggested sequence · Semester 3 | Student-authored study guide

How to use these notes

Search, planning, knowledge representation, uncertainty, machine learning, and responsible AI. Work through the units in order, reproduce each example, then answer the review prompts from memory. This is supplemental practice material, not an official MBU syllabus or assessment. Course sequencing is suggested and should be checked against your current approved syllabus.

MODULE 01 · Agents, states, and search

An agent perceives and acts in an environment. A search problem specifies states, actions, transition model, goal test, and costs. BFS is complete on finite branching unweighted spaces; uniform-cost search handles nonnegative costs.

Worked example

```
A route planner uses intersections as states, roads as actions, and distance or travel time as path cost.
```

Check your understanding

- What defines a search problem?
- Why use a priority queue in uniform-cost search?

MODULE 02 · Heuristics and decision making

A heuristic estimates remaining cost. Admissible heuristics never overestimate optimal cost; consistent heuristics satisfy a triangle-like condition. Decision systems should represent uncertainty and compare actions against explicit objectives.

Worked example

```
Straight-line distance is a lower-bound heuristic for road distance, though it may ignore traffic.
```

Check your understanding

- What does admissible mean?
- How does a heuristic affect search efficiency?

MODULE 03 · Knowledge, learning, and responsibility

Represent facts and relations explicitly when rules are available; use learned models when patterns must be inferred. Evaluate reliability, privacy, bias, misuse, and human oversight, especially where errors affect people.

Worked example

```
A medical decision-support tool needs calibrated evidence, privacy controls, and qualified
human review.
```

Check your understanding

- When can explicit rules be easier to audit?
- Name two responsible-AI evaluation concerns.

Revision checklist

- %¡ Can I explain the main idea without copying the notes?
- %¡ Can I reproduce the example and state its assumptions?
- %¡ Can I test an edge case and describe what should happen?
- %¡ Have I recorded a limitation and checked a primary reference?

Further reading

Use the official documentation and open references linked in the portal. Check the publication/version and license at the source before redistribution. Cite the source and access date in coursework.